Fiscal Year:	FY 2008	**Task Last Updated:**	FY 11/25/2008
PI Name:	Love, Nancy Ph.D.		
Project Title:	Integrated Biotreatment Technology for Nitrogen-Rich Wastewaters in Advanced Life Support Systems		
Division Name:	Human Research		
Program/Discipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES		
Program/Discipline--Element/Subdiscipline:	ADVANCED HUMAN SUPPORT TECHNOLOGIES--Advanced life support		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	nglove@umich.edu	**Fax:**	FY 734-764-4292
PI Organization Type:	UNIVERSITY	**Phone:**	734-764-8495
Organization Name:	Virginia Tech		
PI Address 1:	Environmental Engineering Dept.		
PI Address 2:	418 Durham Hall (0246)		
PI Web Page:			
City:	Blacksburg	**State:**	VA
Zip Code:	24061	**Congressional District:**	9
Comments:	The address above is the address I had when I completed the work. Virginia Tech remains the responsible institution. My current address is: Dept of Civil and Environmental Engineering, University of Michigan, 2340 GG Brown, 2350 Hayward Street, Ann Arbor, MI 48109		
Project Type:	Ground	**Solicitation / Funding Source:**	03-OBPR-01
Start Date:	09/23/2004	**End Date:**	08/31/2008
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: Extension to 8/31/2008 per PI (11/2008) NOTE: Received NCE through 9/30/2007 per PI (11/2008)		
Key Personnel Changes/Previous PI:	The original PI was Barth Smets. Dr. Smets left the University of Connecticut to the Technical University of Denmark (outside the US) before this grant was awarded. Dr. Nancy Love, then of Virginia Tech, took over as PI. Dr. Smets served as a senior technical advisor throughout the project. Dr. Love left Virginia Tech for the University of Michigan during the time that the final report was to be submitted. Between her move and the various unfortunate circumstances that affected many in the Virginia Tech community in 2007, we did not complete the project until early August 2008. The report submitted here is primarily the work of Kevin Gilmore, who recently completed his dissertation topic on the project.		
COI Name (Institution):			
Grant/Contract No.:	NNJ04HD78G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

The goal of this project was to develop a functional, closed-loop water recovery system for Space Shuttle and International Space Station operations. To do so, technologies are required to treat blended urine and humidity condensate wastewaters that contain high concentrations of reduced nitrogen (~ 1.6 g/L as N). Removing this N via conventional treatment means is both resource and cost intensive. This project was proposed to develop, test, and optimize a novel, less resource-intensive, microgravity-compatible, high-capacity nitrogen removal technology termed the oxygen-limited autotrophic nitrification-denitrification (OLAND) process. The OLAND process offers the advantages of effective nitrogen removal to dinitrogen gas (N2), no reliance on external electron donors, and low biomass generation. The novel application for this project involves applying OLAND using a hollow fiber membrane bioreactor (HFMBR), which provides ample support for biofilm growth, permits bubbleless gas addition, enables very high volumetric removal rates, and combines the successfully demonstrated OLAND technology within a HFMBR configuration that permits increased process control. It was originally anticipated that at least 90% of the total Kjeldahl nitrogen in the wastewater will be converted to N2 with the remaining exiting as nitrate-N; this can be removed downstream via biological or physico-chemical polishing.

An OLAND-HFMBR was constructed and assessed using organic carbon-free synthetic wastewater. Because the extent of oxygenation is a key parameter for controlling the metabolic processes that occur in a wastewater treatment system, the first part of the research investigated oxygen transfer characteristics of the HFMBR in clean water conditions and with actively growing biofilm. A mechanistic model for oxygen concentration and flux as a function of length along the non-porous membrane fibers that comprise the HFMBR was developed based on material properties and physical dimensions. This model reflects the diffusion mechanism of non-porous membranes; namely that oxygen follows a sorption-dissolution-diffusion mechanism. This is in contrast to microporous membranes in which oxygen is in the gas phase in the fiber pores up to the membrane surface, resulting in higher biofilm pore liquid dissolved oxygen concentrations. Compared to off-gas oxygen analysis from the HFMBR while in operation with biofilm growing, the model overpredicted mass transfer by a factor of approximately 1.3. This was in contrast to empirical mass transfer coefficient-based methods, which were determined using either bulk aqueous phase dissolved oxygen (DO) concentration or the DO concentration at the membrane-liquid interface, measured with oxygen microsensors. The mass transfer coefficient determined with the DO measured at the interface was the best predictor of actual oxygen transfer under biofilm conditions, while the bulk liquid coefficient underpredicted by a factor of 3. The mechanistic model exhibited sensitivity to parameters such as the initial lumen oxygen concentration (at the entry to the fiber) and the diffusion coefficient and partitioning coefficients of oxygen in the silicone membrane material. The mechanistic model has several advantages over empirical-based methods. Namely, it does not require experimental determination of KL, it is relatively simple to solve without the use of advanced mathematical software, and it is based upon selection of the membrane-biofilm interfacial DO concentration. The last of these is of particular importance when designing and operating HFMBR systems with redox (aerobic/anoxic/anaerobic) stratification, because the DO concentration will determine the nature of the microenvironments, the microorganisms present, and the metabolisms that occur.

During the second phase of the research, the coupling of two autotrophic metabolisms, partial nitrification to nitrite (nitritation) and anaerobic ammonium oxidation, was demonstrated in a single HFMBR. The system successfully treated a high-strength nitrogen wastewater intended to mimic a urine stream from such sources as extended space missions. For the last 250 days of operation, operating with an average oxygen to ammonia flux (JO2/JNH4+) of 3.0 resulted in an average nitrogen removal of 74%, with no external organic carbon added. Control of nitrite-oxidizing bacteria (NOB) presented a challenge that was addressed by maintaining the JO2/JNH4+ below the stoichiometric threshold for complete nitrification to nitrate (4.57 g O2 / g NH4+). Of particular concern for application of this technology for space flight, we found that the DO-limiting condition resulted in formation of harmful gaseous emissions of nitrogen oxides (NO, N2O), which could not be prevented by short-term control strategies. Controlling JO2/JNH4+ prevented NOB proliferation long enough to allow an anaerobic ammonia-oxidizing bacteria (AnaerAOB) population to develop and be retained for >250 days. Addition of a supplemental nutrient solution may have contributed to the growth of AnaerAOB by overcoming a possible micronutrient deficiency. Disappearance of the gaseous nitrogen oxide emissions coincided with the onset of anaerobic ammonium oxidation, demonstrating a benefit of coupling these two autotrophic metabolisms in one reactor. Obvious differences in biofilm density were evident across the biofilm depth, with a region of low density in the middle of the biofilm, suggesting that low cell density or exocellular polymeric substances were primarily present in this region. Microbial community analysis using fluorescence in situ hybridization (FISH) did not reveal consistent trends with respect to length along the fibers, but radial stratification of aerobic ammonia-oxidizing bacteria (AerAOB), NOB, and AnaerAOB were visible in biofilm section samples. AerAOB were largely found in the first 25% of the biofilm near the membrane, AnaerAOB were found in the outer 30%, and NOB were found most often in the mid-depth region of the biofilm. This community structure demonstrates the importance of oxygen availability as a determinant of how microbial groups spatially distribute within an HFMBR biofilm.

The combination of these two aspects of the research, predictive oxygen transfer capability and the effect of oxygen control on performance and populations, provides a foundation for future application of HFMBR technology to a broad range of earth-based and planetary or long-term space flight wastewater treatment scenarios. However, the onset of hazardous nitrogen oxide gases during the long acclimation phase of the reactor (removed later when anaerobic ammonia oxidation was initiated) presents a challenge when considering how one would apply this technology to long term space flight or planetary base stations. Nevertheless, the successful demonstration of this technology in a ground-based research project paves the way for further testing and validation in simulated microgravity environments on the ground as well as testing and demonstration of a prototype unit in space. Furthermore, the results are pertinent to significant applications on Earth, where use of anaerobic treatment technologies for sustainable waste management are increasing in frequency. Such applications generate strong ammonia-ladened waste streams that may benefit from being treated by an OLAND-HFMBR application.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The technology has application to similar high ammonia waste systems on Earth, such as anaerobic treatment of agricultural wastes, domestic sludge processing waters, etc. We expect increased use of anaerobic treatment technologies as the need for sustainable treatment strategies increases. Furthermore, we believe that the outcomes show that the slow startup associated with the technology may make it inappropriate for use for flight-based applications. It may be appropriate for planetary applications, including on Earth, if the startup time can be reduced.

Task Progress:	An OLAND-HFMBR was constructed and assessed using organic carbon-free synthetic wastewater. Because the extent of oxygenation is a key parameter for controlling the metabolic processes that occur in a wastewater treatment system, the first part of the research investigated oxygen transfer characteristics of the HFMBR in clean water conditions and with actively growing biofilm. A mechanistic model for oxygen concentration and flux as a function of length along the non-porous membrane fibers that comprise the HFMBR was developed based on material properties and physical dimensions. This model reflects the diffusion mechanism of non-porous membranes; namely that oxygen follows a sorption-dissolution-diffusion mechanism. This is in contrast to microporous membranes in which oxygen is in the gas phase in the fiber pores up to the membrane surface, resulting in higher biofilm pore liquid dissolved oxygen concentrations. Compared to off-gas oxygen analysis from the HFMBR while in operation with biofilm growing, the model overpredicted mass transfer by a factor of approximately 1.3. This was in contrast to empirical mass transfer coefficient-based methods, which were determined using either bulk aqueous phase dissolved oxygen (DO) concentration or the DO concentration at the membrane-liquid interface, measured with oxygen microsensors. The mass transfer coefficient determined with the DO measured at the interface was the best predictor of actual oxygen transfer under biofilm conditions, while the bulk liquid coefficient underpredicted by a factor of 3. The mechanistic model exhibited sensitivity to parameters such as the initial lumen oxygen concentration (at the entry to the fiber) and the diffusion coefficient and partitioning coefficients of oxygen in the silicone membrane material. The mechanistic model has several advantages over empirical-based methods. Namely, it does not require experimental determination of KL, it is relatively simple to solve without the use of advanced mathematical software, and it is based upon selection of the membrane-biofilm interfacial DO concentration. The last of these is of particular importance when designing and operating HFMBR systems with redox (aerobic/anoxic/anaerobic) stratification, because the DO concentration will determine the nature of the microenvironments, the microorganisms present, and the metabolisms that occur. During the second phase of the research, the coupling of two autotrophic metabolisms, partial nitrification to nitrite (nitritation) and anaerobic ammonium oxidation, was demonstrated in a single HFMBR. The system successfully treated a high-strength nitrogen wastewater intended to mimic a urine stream from such sources as extended space missions. For the last 250 days of operation, operating with an average oxygen to ammonia flux (JO2/JNH4+) of 3.0 resulted in an average nitrogen removal of 74%, with no external organic carbon added. Control of nitrite-oxidizing bacteria (NOB) presented a challenge that was addressed by maintaining the JO2/JNH4+ below the stoichiometric threshold for complete nitrification to nitrate (4.57 g O2 / g NH4+). Of particular concern for application of this technology for space flight, we found that the DO-limiting condition resulted in formation of harmful gaseous emissions of nitrogen oxides (NO, N2O), which could not be prevented by short-term control strategies. Controlling JO2/JNH4+ prevented NOB proliferation long enough to allow an anaerobic ammonia-oxidizing bacteria (AnaerAOB) population to develop and be retained for >250 days. Addition of a supplemental nutrient solution may have contributed to the growth of AnaerAOB by overcoming a possible micronutrient deficiency. Disappearance of the gaseous nitrogen oxide emissions coincided with the onset of anaerobic ammonium oxidation, demonstrating a benefit of coupling these two autotrophic metabolisms in one reactor. Obvious differences in biofilm density were evident across the biofilm depth, with a region of low density in the middle of the biofilm, suggesting that low cell density or exocellular polymeric substances were primarily present in this region. Microbial community analysis using fluorescence in situ hybridization (FISH) did not reveal consistent trends with respect to length along the fibers, but radial stratification of aerobic ammonia-oxidizing bacteria (AerAOB), NOB, and AnaerAOB were visible in biofilm section samples. AerAOB were largely found in the first 25% of the biofilm near the membrane, AnaerAOB were found in the outer 30%, and NOB were found most often in the mid-depth region of the biofilm. This community structure demonstrates the importance of oxygen availability as a determinant of how microbial groups spatially distribute within an HFMBR biofilm. The combination of these two aspects of the research, predictive oxygen transfer capability and the effect of oxygen control on performance and populations, provides a foundation for future application of HFMBR technology to a broad range of earth-based and planetary or long-term space flight wastewater treatment scenarios. However, the onset of hazardous nitrogen oxide gases during the long acclimation phase of the reactor (removed later when anaerobic ammonia oxidation was initiated) presents a challenge when considering how one would apply this technology to long term space flight or planetary base stations. Nevertheless, the successful demonstration of this technology in a ground-based research project paves the way for further testing and validation in simulated microgravity environments on the ground as well as testing and demonstration of a prototype unit in space. Furthermore, the results are pertinent to significant applications on Earth, where use of anaerobic treatment technologies for sustainable waste management are increasing in frequency. Such applications generate strong ammonia-ladened waste streams that may benefit from being treated by an OLAND-HFMBR application.
Bibliography Type:	Description: (Last Updated: 03/23/2017)
Articles in Other Journals or Periodicals	Gilmore KR, Terada A, Smets BF, Love NG. "Controlling population dynamics and nitrogen removal performance in hollow fiber membrane-aerated biofilm reactors." Newsletter of the IWA Specialist Group on Activated Sludge Populations Dynamics, May 2007. , May-2007
Articles in Peer-reviewed Journals	Gilmore KR, Little JC, Smets BF, Love NG. "Oxygen transfer model for a flow-through hollow-fiber membrane biofilm reactor." Journal of Environmental Engineering. 2009 Sep;135(0):806-14. 2008. http://dx.doi.org/10.1061/(ASCE)EE.1943-7870.0000035 , Sep-2009
Articles in Peer-reviewed Journals	Gilmore KR, Terada A, Smets BF, Love NG, Garland JL. "Autotrophic nitrogen removal in a membrane-aerated biofilm reactor under continuous aeration: a demonstration." Environmental Engineering Science. 2013 Jan;30(1):38-45. http://dx.doi.org/10.1089/ees.2012.0222 , Jan-2013
Papers from Meeting Proceedings	Capuno RE, Love NG, Smets BF. "Mathematical modeling of start-up scenarios for nitrogen removal via a nitritation:anaerobic ammonia oxidation-coupled biofilm in a hollow fiber membrane bioreactor." International Water Association Biofilm Systems VI, Amsterdam, The Netherlands, September 24-27, 2006. Proceedings of the International Water Association Biofilm Systems VI, Amsterdam, The Netherlands, September 24-27, 2006 , Sep-2006
Papers from Meeting Proceedings	Gilmore KR, Love NG, Smets BF, Terada A, Garland J. "Nitrifier and anammox population dynamics in an autotrophic nitrogen removal membrane biofilm reactor." Podium presentation. Water Environment Federation 81st Annual Conference and Exposition, Chicago, IL, October 19-22, 2008. Proceedings of the Water Environment Federation 81st Annual Conference and Exposition, October 2008. , Oct-2008

Papers from Meeting Proceedings	Gilmore KR, Capuno RE Jr, Love NG, Smets BF. "Anaerobic stabilization of early planetary base ersatz wastewater formulation." 36th International Conference on Environmental Systems, SAE, Norfolk, VA , July 2006. Society of Automotive Engineers (SAE) Technical Paper Series 2006-01-2255, July 2006. , Jul-2006
Papers from Meeting Proceedings	Gilore KR, Smets BF, Garland JL, Terada A, Love NG. "Controlling gaseous nitrogen oxide emissions and nitrogen removal performance in hollow fiber membrane-aerated biofilm reactors." WEF Membrane Technology 2008 Conference, Atlanta, Georgia, January 27-30, 2008. Proceedings of the WEF Membrane Technology 2008 Conference, January 2008. , Jan-2008
Papers from Meeting Proceedings	Gilmore KR, Little JC, Love NG, Smets BF. "Oxygen mass transfer in a flow-through hollow-fiber membrane aeration reactor." 80th Annual Water Environment Federation Technical Exhibition and Conference (WEFTEC), San Diego, California, October 13-17, 2007. Proceedings of the 80th Annual Water Environment Federation Technical Exhibition and Conference (WEFTEC), October 2007. , Oct-2007